

CHAPTER – 17

DETERMINANTS AND MATRICES

Properties of Determinants, Cramer's Rule and Inverse of Matrix

1. Introduction to Determinants

Definition

A determinant is a special number associated with a square matrix.

For matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

the determinant is written as:

$$|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

and its value is:

$$|A| = ad - bc$$

2. Determinant of Order 3

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

$$= a(ei - fh) - b(di - fg) + c(dh - eg)$$

3. Properties of Determinants

Property 1: Interchange of Rows

Property

If two rows (or columns) are interchanged, determinant changes sign.

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = -\begin{vmatrix} c & d \\ a & b \end{vmatrix}$$

Property 2: Equal Rows

If two rows or columns are equal, determinant becomes zero.

$$\begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} = 0$$

Property 3: Multiplying a Row

If all elements of a row are multiplied by k , determinant is multiplied by k .

$$\begin{vmatrix} ka & kb \\ c & d \end{vmatrix} = k \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

Property 4: Zero Row

If all elements of a row or column are zero, determinant is zero.

$$\begin{vmatrix} 0 & 0 \\ a & b \end{vmatrix} = 0$$

Property 5: Transpose Property

Determinant remains same after transpose.

$$|A| = |A^T|$$

4. Examples on Determinants

Example 1

Find:

$$\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix}$$

Solution

$$= (2)(5) - (3)(4)$$

$$= 10 - 12$$

$$= -2$$

$$\boxed{-2}$$

Example 2

Evaluate:

$$\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{vmatrix}$$

Solution

$$= 1[(1)(0) - (4)(6)] - 2[(0)(0) - (4)(5)] + 3[(0)(6) - (1)(5)]$$

$$= -24 + 40 - 15$$

$$= 1$$

$$\boxed{1}$$

5. Cramer's Rule

Cramer's Rule

Cramer's Rule is used to solve linear equations using determinants.

For equations:

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Define:

$$\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

$$\Delta_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}$$

$$\Delta_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$$

Then:

$$x = \frac{\Delta_x}{\Delta}$$

$$y = \frac{\Delta_y}{\Delta}$$

6. Solving Equations by Cramer's Rule

Example 3

Solve:

$$2x + 3y = 8$$

$$x - y = 1$$

Solution

$$\Delta = \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix}$$

$$= (2)(-1) - (3)(1)$$

$$= -2 - 3 = -5$$

$$\begin{aligned}\Delta_x &= \begin{vmatrix} 8 & 3 \\ 1 & -1 \end{vmatrix} \\ &= (8)(-1) - (3)(1) \\ &= -8 - 3 = -11\end{aligned}$$

$$\begin{aligned}\Delta_y &= \begin{vmatrix} 2 & 8 \\ 1 & 1 \end{vmatrix} \\ &= (2)(1) - (8)(1) \\ &= 2 - 8 = -6\end{aligned}$$

Now:

$$\begin{aligned}x &= \frac{-11}{-5} = \frac{11}{5} \\ y &= \frac{-6}{-5} = \frac{6}{5}\end{aligned}$$

$$\boxed{x = \frac{11}{5}, \quad y = \frac{6}{5}}$$

7. Inverse of a Matrix

Definition

For a square matrix A ,
if there exists a matrix A^{-1} such that:

$$AA^{-1} = I$$

then A^{-1} is called inverse of matrix A .

8. Formula for Inverse of 2×2 Matrix

For:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Inverse exists only if:

$$|A| = ad - bc \neq 0$$

Formula:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

9. Examples on Inverse of Matrix

Example 4

Find inverse of:

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

Solution

Find determinant:

$$|A| = (2)(4) - (3)(1)$$

$$= 8 - 3 = 5$$

Since determinant is non-zero, inverse exists.

Using formula:

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{4}{5} & -\frac{3}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix}$$

10. Verification of Inverse

Example 5

Verify inverse for:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

Multiply:

$$\begin{aligned} AA^{-1} &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

Hence verified.

11. Advanced Examples

Example 6

Evaluate determinant:

$$\begin{vmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 4 & 1 & 8 \end{vmatrix}$$

Solution

Expand along first row:

$$\begin{aligned} &= 2 \begin{vmatrix} 0 & 2 \\ 1 & 8 \end{vmatrix} - 1 \begin{vmatrix} 1 & 2 \\ 4 & 8 \end{vmatrix} + 3 \begin{vmatrix} 1 & 0 \\ 4 & 1 \end{vmatrix} \\ &= 2(0 - 2) - 1(8 - 8) + 3(1 - 0) \end{aligned}$$

$$= -4 + 0 + 3$$

$$= -1$$

$$\boxed{-1}$$

Example 7

Solve using Cramer's Rule:

$$3x + 2y = 7$$

$$2x - y = 4$$

Solution

$$\begin{aligned}\Delta &= \begin{vmatrix} 3 & 2 \\ 2 & -1 \end{vmatrix} \\ &= -3 - 4 = -7\end{aligned}$$

$$\begin{aligned}\Delta_x &= \begin{vmatrix} 7 & 2 \\ 4 & -1 \end{vmatrix} \\ &= -7 - 8 = -15\end{aligned}$$

$$\begin{aligned}\Delta_y &= \begin{vmatrix} 3 & 7 \\ 2 & 4 \end{vmatrix} \\ &= 12 - 14 = -2\end{aligned}$$

Thus:

$$x = \frac{-15}{-7} = \frac{15}{7}$$

$$y = \frac{-2}{-7} = \frac{2}{7}$$

$$\boxed{x = \frac{15}{7}, \quad y = \frac{2}{7}}$$

12. Important Formula Summary

Concept	Formula
Determinant of 2×2	$ad - bc$
Cramer's Rule	$x = \frac{\Delta_x}{\Delta}, \quad y = \frac{\Delta_y}{\Delta}$
Inverse of Matrix	$A^{-1} = \frac{1}{ A } \text{Adj}(A)$
Inverse of 2×2	$\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

13. Practice Questions

1. Find:

$$\begin{vmatrix} 3 & 4 \\ 2 & 5 \end{vmatrix}$$

2. Evaluate:

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{vmatrix}$$

3. Solve using Cramer's Rule:

$$x + y = 5$$

$$2x - y = 1$$

4. Find inverse of:

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$$

5. Verify:

$$AA^{-1} = I$$

for:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

6. Solve:

$$3x + 4y = 10$$

$$2x - y = 5$$

7. Find determinant:

$$\begin{vmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 4 & 1 & 0 \end{vmatrix}$$

8. Find inverse of:

$$\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$$

9. Determine whether inverse exists for:

$$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

10. Solve:

$$x + 2y = 7$$

$$3x - y = 5$$

14. Answer Key

1.

$$7$$

2.

$$-2$$

3.

$$x = 2, \quad y = 3$$

4.

$$\begin{bmatrix} 5 & -2 \\ -3 & 1 \end{bmatrix}$$

5. Verified

6.

$$x = \frac{30}{11}, \quad y = \frac{5}{11}$$

7.

$$11$$

8.

$$\frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

9. Inverse does not exist

10.

$$x = \frac{17}{7}, \quad y = \frac{16}{7}$$